

Course Information

Course Number: ISEN 427/DAEN 427
Course Title: Decision and Risk Analysis
Section: 500
Time: TR 9:35-10:50 AM
Location: [ETB](#) 1013
Credit Hours: 3 (3 hours of lecture per week)

Instructor & TA/Grader

Instructor: Alexander Abuabara
Email: abuabara@tamu.edu
Zoom: <https://tamu.zoom.us/my/abuabara>
Office: [ETB](#) 4013
Office Hours: W 1-2:30 PM or by appointment
TA: TBA

Course Description

Overview of the state of the art in descriptive and prescriptive theories of decision making under uncertainty with emphasis on the ways in which human decisions depart from normative models of rationality; analytical foundations stemming from several disciplines, economics, psychology, management science; application in engineering systems will be considered.

Learning Outcomes

At the end of this course, the student should be able to:

- Draw on concepts from economics, psychology, and management science to analyze decision-making processes.
- Understand Bayesian approach to perception and decisions under uncertainty.
- Implement computational models for Bayesian inference.

Course Prerequisites

Grade of C or better in ISEN 310, DAEN 321, or STAT 212

Cross Listing

DAEN 427/ISEN 427

Course Expectations

You are expected to:

- Prepare by completing assigned readings before class,
- Participate in discussions and problem-solving activities,
- Submit assignments and projects on time, demonstrating responsibility and clear communication, and
- Maintain academic integrity in all submitted work.

Recommended Textbooks

- [Bayesian Models of Perception and Action: An Introduction. W. J. Ma, K. Kording, D. Goldreich \(2022\).](#) [BM]
- [An Introduction to Bayesian Inference, Methods and Computation. N. Heard \(2021\).](#) [IBI]
- [A Course in Behavioral Economics. E. Angner \(2021\).](#) [BE]
- [Bayesian Modeling and Computation in Python. O. A. Martin, R. Kumar, J. Lao \(2021\).](#) [BCP]

Grading Policy

- Graded items and weights:
 - **Assignments/Participation (10%)**
 - **Mid-term exam (x2, 30% each)**
 - **Final project: project report (25%), project presentation (5%)**
- Grades will be maintained in the Gradebook on Canvas.
- Grades will be calculated by totaling the points you earn and comparing them to the max. possible points.
- Grades assigned are A: 90%–100%, B: 80%–89.9%, C: 70%–79.9%, D: 60%–69.9% and F for less than 60%.

Graded Class Participation – In class assignments/quizzes

Graded Attendance – None

Grades for Stacked Course – N/A

Grading Policy Changes – No changes after the second class; report issues promptly.

Late Work Policy

Late work will not be accepted except for approved university excused absences. Late work will be assigned a score of zero. Disagreements regarding a grade received on any graded material must be discussed within one week of the return of the graded material. No grade will be changed beyond this limit. If you request a regrade, the entire submission will be regarded.

Course Assignments

Assignments will be distributed after each topic as appropriate. Some will be done in class (including weekly quizzes). Homework assignments will allow one week for completion. While you may discuss assignments with classmates, all work must be completed independently and submitted individually unless otherwise specified. Assignments help you practice class concepts: they are not graded but you'll earn participation points for clear, on-time work.

Exams are cumulative and may cover material from any part of the course to date. Typically, an exam will examine the same concepts, just in different way than the assignments. Exams will be held in class on Sep 25 and Oct 30 (TBC).

Projects will be defined following the fall break. This is an individual project where you'll put core concepts into practice. Projects are due in class on Dec 4, with presentations on the same day (TBC).

Further instructions for assignments, exams, and project will be provided in class.

Attendance & Class Participation

Attendance is required for exams and presentations. You are expected to attend all class lectures except for university excused absences. The university rule regarding excused absences can be found at <https://student-rules.tamu.edu/rule07>. Note: you can ONLY complete the in-class activities if you attended the class on the day the activity was given. Completing in-class activities without attending the class (or notifying the instructor in advance) will be considered academic misconduct.

Course Schedule

Tentative (subject to adjustment): dates will not be changed without written notification to all students in the course (notice via email & Canvas), and additional readings may be assigned on Canvas as necessary.

Lecture	Topic	Readings	Graded Items
1	Intro. Graphical modeling. Perception as inference. Bayesian inference.	BM 1-2, IBI 3	Midterm exam 1 (Set 25, in class)
2	Uncertainty and confidence. Bayesian inference under measurement noise.	BM 3-4	
3	Predicting behaviors, estimation (MLE, MAP).	BM 4	
4	Cue combination, learning as inference.	BM 5-6	
5	Computational aspects of Bayes inference (MCMC).	IBI 5, BCP 1-2	Midterm exam 2 (Oct 30, in class)
6	Discrimination and detection. Binary classification.	BM 7, IBI 7	
7	Bayesian linear regression.	IBI 8, BCP 3	
8	Combining inference with utility.	BM 9, BE 6	
9	Rational choice under risk.	BM 9, BCP 4	Project (Dec 4, in class)
10	Some paradoxes, prospect theory.	BE 7	
11	Expectation maximization algorithm.	BE 7	
12	Variational inference. Bayesian inference with neural networks.	BMCP 14	

Important Dates (<https://registrar.tamu.edu/Academic-Calendar>)

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|-------------|--|-------------|-----------------------------------|
| • Aug 25 | First day of Fall semester classes | • Nov 26 | Reading day, no classes |
| • Aug 29 | Last day to add/drop courses for Fall semester | • Nov 27-28 | Thanksgiving holiday, no classes |
| • Sep 1 | Labor day, no classes | • Dec 8 | Last day of Fall semester classes |
| • Oct 13-14 | Fall break, no classes | • Dec 9-10 | Reading days, no classes |

Tips

(i) You all are important to me: include the course number in email subject so I can prioritize it. (ii) Every class matter and each build on the next: stay engaged, keep up, ask questions. (iii) Learning takes steady effort, start early, not last-minute cramming. (iv) Engineers turn data into meaningful information, practice here, get prepared. (v) Clear, well-organized work is as important as the content itself. (vi) Use others' ideas to build your own but always credit your sources: good scholarship means giving credit where it's due. (vii) No audio or video recording without permission; it's a conduct violation otherwise.

University Policies

Attendance Policy

The university views class attendance and participation as an individual student responsibility. Students are expected to attend class and to complete all assignments. Please refer to [Student Rule 7](#) in its entirety for information about excused absences, including definitions, and related documentation and timelines.

Student Observances for Religious Holy Days

In accordance with Texas Education Code §51.911(b) and Texas A&M Student Rule 7: Attendance, students shall be excused from attending classes or other required activities, including examinations, for the observance of a religious holy day, including travel for that purpose. For more information about excused absences due to religious holy days, please visit the [Faculty Affairs website](#) (under Other University Guidelines).

Makeup Work Policy

Students will be excused from attending class on the day of a graded activity or when attendance contributes to a student's grade, for the reasons stated in Student Rule 7, or other reason deemed appropriate by the instructor. Please refer to [Student Rule 7](#) in its entirety for information about makeup work, including definitions, and related documentation and timelines.

Absences related to Title IX of the Education Amendments of 1972 may necessitate a period of more than 30 days for make-up work, and the timeframe for make-up work should be agreed upon by the student and instructor" ([Student Rule 7, Section 7.4.1](#)).

"The instructor is under no obligation to provide an opportunity for the student to make up work missed because of an unexcused absence" ([Student Rule 7, Section 7.4.2](#)).

Students who request an excused absence are expected to uphold the Aggie Honor Code and Student Conduct Code. (See [Student Rule 24](#).)

Academic Integrity Statement and Policy

"An Aggie does not lie, cheat or steal, or tolerate those who do."

"Texas A&M University students are responsible for authenticating all work submitted to an instructor. If asked, students must be able to produce proof that the item submitted is indeed the work of that student. Students must keep appropriate records at all times. The inability to authenticate one's work, should the instructor request it, may be sufficient grounds to initiate an academic misconduct case" ([Section 20.1.2.3, Student Rule 20](#)).

You can learn more about the Aggie Honor System Office Rules and Procedures, academic integrity, and your rights and responsibilities at aggiehonor.tamu.edu.

Plagiarism Policy

According to the Texas A&M University Definitions of Academic Misconduct, plagiarism is the appropriation of another person's ideas, processes, results or words without giving appropriate credit (aggiehonor.tamu.edu). You should credit your use of anyone else's words, graphic images, or ideas using standard citation styles. AI text generators (ChatGPT, Google Bard, etc.) should not be used for any work for this class without explicit permission of the instructor and appropriate attribution. This includes, but is not limited to:

- Creating or revising drafts
- Editing your work
- Reviewing a peer's work

This excludes pre-existing software additions such as spelling and grammar checkers, which are acceptable.

Such use of AI text generators in this manner could be considered plagiarism and cheating according to Student Rule 20. More information may also be found at <https://aggiehonor.tamu.edu> or you may contact your instructor if you have questions.

Americans with Disabilities Act (ADA) Policy

Texas A&M University is committed to providing equitable access to learning opportunities for all students. If you experience barriers to your education due to a disability or think you may have a disability, please contact the Disability Resources office on your campus (resources listed below) Disabilities may include, but are not limited to attentional, learning, mental health,

sensory, physical, or chronic health conditions. All students are encouraged to discuss their disability related needs with Disability Resources and their instructors as soon as possible.

Disability Resources is in the Student Services Building or at (979) 845-1637 or visit disability.tamu.edu.

Title IX and Statement on Limits to Confidentiality

Texas A&M University is committed to fostering a learning environment that is safe and productive for all. University policies and federal and state laws prohibit gender-based discrimination and sexual harassment, including sexual assault, sexual exploitation, domestic violence, dating violence, and stalking.

With the exception of some medical and mental health providers, all university employees (including full and part-time faculty, staff, paid graduate assistants, student workers, etc.) are Mandatory Reporters and must report to the Title IX Office if the employee experiences, observes, or becomes aware of an incident that meets the following conditions (see [University Rule 08.01.01.M1](#)):

- The incident is reasonably believed to be discrimination or harassment.
- The incident is alleged to have been committed by or against a person who, at the time of the incident, was (1) a student enrolled at the University or (2) an employee of the University.

Mandatory Reporters must file a report regardless of how the information comes to their attention – including but not limited to face-to-face conversations, a written class assignment or paper, class discussion, email, text, or social media post. Although Mandatory Reporters must file a report, in most instances, a person who is subjected to the alleged conduct will be able to control how the report is handled, including whether to pursue a formal investigation. The University's goal is to make sure you are aware of the range of options available to you and to ensure access to the resources you need.

Students wishing to discuss concerns related to mental and/or physical health in a confidential setting are encouraged to make an appointment with [University Health Services](#) or download the [TELUS Health Student Support app](#) for 24/7 access to professional counseling in multiple languages. Walk-in services for urgent, non-emergency needs are available during normal business hours at University Health Services locations; call 979.458.4584 for details.

Students can learn more about filing a report, accessing supportive resources, and navigating the Title IX investigation and resolution process on the University's [Title IX webpage](#).

Statement on Mental Health and Wellness

Texas A&M University recognizes that mental health and wellness are critical factors influencing a student's academic success and overall wellbeing. Students are encouraged to engage in healthy self-care practices by utilizing the resources and services available through [University Health Services](#). The [TELUS Health Student Support](#) app provides access to professional counseling in multiple languages anytime, anywhere by phone or chat, and the 988 Suicide & Crisis Lifeline offers 24-hour emergency support at 988 or 988lifeline.org.

Pregnancy Accommodations

Texas A&M provides reasonable accommodations to students due to pregnancy and/or related conditions, such as childbirth, recovery and lactation. Students should contact the University's [Pregnancy Coordinator](#) as soon as they become aware of the need for accommodation. Depending on the circumstances, accommodations could include extended time to complete assignments or exams, changes in course sequence, or modifications to the physical classroom environment. Texas A&M will also allow a voluntary leave of absence, ensure the availability of lactation space, and maintain grievance procedures to provide for the prompt and equitable resolution of complaints of sex discrimination. For information regarding pregnancy accommodations, email TIX.Pregnancy@tamu.edu.